Consensus Control for Multi-Agent Systems under Switching Topologies via State-Dependent Dwell-Time Switching

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***Abstract*—This paper investigates the consensus problem for switched multi-agent systems with multiple candidate directed communication topologies. A consensus-orthogonal decomposition is first introduced to eliminate the redundant consensus component and transform the original multi-agent system into a reduced-order switched disagreement system. A state-dependent dwell-time switching law is then designed via Lyapunov-Metzler inequalities to stabilize the disagreement error system. Under the jointly strongly connected topology condition, the Lyapunov-Metzler inequalities are proved to be feasible for every prescribed minimum dwell time below an explicitly derived positive upper bound. It is shown that, the disagreement error converges to zero and global consensus is achieved, provided that the union of the candidate communication graphs is strongly connected. A numerical example is provided to illustrate the effectiveness of the proposed approach.**

***Index Terms*—Multi-agent consensus, switching topology, joint strong connectivity, Lyapunov-Metzler inequality, state-dependent dwell-time.**

I. Introduction

Consensus control is a fundamental problem in multi-agent systems, where a group of agents is required to reach agreement through local information exchange over communication networks [1], [2]. Existing studies have extended consensus control to second-order systems with velocity and input constraints and finite-time protocols with actuator saturation [3], [4]. And resilient consensus for higher order multi-agent networks has also been investigated in [5].

In graph-based consensus analysis, connectivity of the communication topology plays a fundamental role. For undirected networks, connectivity is a standard condition for consensus [6], whereas directed networks are typically required to be strongly connected or to contain a directed spanning tree [7], [8]. In practical networked systems, however, communication topologies may switch due to link failures, sensing limitations, communication constraints, or intermittent information exchange. Representative studies have investigated consensus tracking of multi-agent systems with Lipschitz-type node dynamics under switching topologies [9] and developed multiple-Lyapunov-function approaches to tracking control of multi-agent systems with switching topologies [10]. Although a single candidate topology may be disconnected and fail to guarantee global consensus independently, the union of several candidate topologies may be strongly connected and provide sufficient information paths. Existing studies have shown that consensus can still be achieved under jointly connected switching topologies even when each individual topology is insufficient [11], [12].

An effective approach to consensus analysis is to separate the consensus and disagreement components. Through an orthogonal transformation, the original consensus problem can be converted into the stability problem of a reduced-order disagreement system [13]. The resulting model removes the redundant consensus direction and permits the direct application of switched-system stability methods.

The reduced disagreement dynamics constitute a switched system whose individual subsystem matrices may be unstable or marginally stable. Classical switched-system theory has shown that a suitable switching law can stabilize the overall system without requiring every subsystem to be Hurwitz [14]–[16]. More recent studies have developed mode-dependent dwell-time and multiple-Lyapunov-function methods for switched systems containing unstable modes [17], [18]. These results motivate treating consensus under individually disconnected topologies as a switched-system stabilization problem.

This paper addresses the consensus problem in multi-agent systems with multiple candidate directed communication topologies. Each candidate topology is disconnected, whereas the union of all candidate topologies is strongly connected. Motivated by the state-dependent dwell-time framework in [19], a Lyapunov-Metzler-based switching mechanism is applied to the reduced-order disagreement dynamics. An orthogonal decomposition removes the consensus direction, and the proposed switching law drives the disagreement state to zero, thereby achieving global consensus.

The main contributions are summarized as follows:

- A consensus-orthogonal decomposition is introduced to transform the original consensus problem into stabilization of a reduced-order switched disagreement system.

- A state-dependent switching law based on finite-dwell-time Lyapunov-Metzler inequalities is developed. The law enforces a prescribed minimum dwell time without requiring the individual subsystem matrices to be Hurwitz.
- A constructive feasibility analysis is established through a Hurwitz convex combination, a specially constructed Metzler matrix, and a lifted Lyapunov operator. An explicit sufficient upper bound on the prescribed dwell time is consequently derived.

The remainder of this paper is organized as follows. Section II presents the graph preliminaries and reduced-order disagreement dynamics. Section III introduces the state-dependent dwell-time switching law. Section IV establishes the feasibility of the Lyapunov-Metzler inequalities. Section V presents numerical simulations, and Section VI concludes the paper.

Notations: $\mathbb{R}^n$ denotes the n-dimensional Euclidean space. The vector $\mathbf{1}_N \in \mathbb{R}^N$ stands for the N-dimensional all-ones vector. $I_n \in \mathbb{R}^{n\times n}$ is the n-order identity matrix, and $\mathbf{0}_n \in \mathbb{R}^n$ denotes the n-dimensional zero vector. For a symmetric matrix P, $P \succ 0$ and $P \prec 0$ denote positive and negative definiteness. The symbol $\otimes$ denotes the Kronecker product. For square matrices A and B, $A \oplus B = A \otimes I + I \otimes B$ denotes the Kronecker sum. For vectors, $\|\cdot\|_2$ is the Euclidean norm; for matrices, it is the induced spectral norm. For a square matrix A, $\mathrm{spec}(A)$ denotes its spectrum. The notation $\mathrm{col}\{\cdot\}$ denotes column stacking. For a matrix $X \in \mathbb{R}^{m\times n}$, $\mathrm{vec}(X) \in \mathbb{R}^{mn}$ denotes the column-wise vectorization obtained by stacking the columns of X. The class $\mathcal{M}$ contains Metzler matrices $\Pi = [\pi_{ij}] \in \mathbb{R}^{M\times M}$ satisfying $\pi_{ij} \geq 0$ for $i \neq j$ and $\Pi \mathbf{1}_M = \mathbf{0}_M$. The set of real symmetric matrices of dimension r is denoted by $\mathbb{S}^r$, and $\mathbb{S}^r_+ = \{Z \in \mathbb{S}^r : Z \succeq 0\}$ denotes the positive-semidefinite cone. The notation $(\mathbb{S}^r_+)^M$ denotes its M-fold Cartesian product. For a matrix Z, $\ker(Z)$ denotes its null space. For a closed convex set $\mathcal{K}$ and $Z \in \mathcal{K}$, $T_{\mathcal{K}}(Z)$ denotes the tangent cone to $\mathcal{K}$ at Z.

II. Preliminaries and Problem Formulation

Consider N agents with states $x_i(t) \in \mathbb{R}^d$, $i \in \mathcal{V}$, where $\mathcal{V} = \{1,\ldots,N\}$ is the agent set. Define $x(t) = \mathrm{col}\{x_1(t),\ldots,x_N(t)\} \in \mathbb{R}^{Nd}$. The communication network of these N agents switches among M candidate directed graphs $\mathcal{G}_p = (\mathcal{V},\mathcal{E}_p), p \in \Omega = \{1,\ldots,M\}$, where $\mathcal{E}_p \subseteq \mathcal{V}\times\mathcal{V}$ is the directed edge set. Define $\mathcal{A}_p = [a^p_{ij}] \in \mathbb{R}^{N\times N}$ as the weighted adjacency matrix. A directed edge $(j,i) \in \mathcal{E}_p$ means that agent i can receive information from agent j. Accordingly, $a^p_{ij} > 0$ if $(j,i) \in \mathcal{E}_p$, while $a^p_{ij} = 0$ indicates the absence of the directed edge from agent j to agent i. The associated Laplacian matrix $L_p = [l^p_{ij}]$ satisfies $l^p_{ii} = \sum_{j\neq i} a^p_{ij}$, $l^p_{ij} = -a^p_{ij}$, $i \neq j$, which yields $L_p \mathbf{1}_N = \mathbf{0}_N$, $p = 1,\ldots,M$. A directed path from agent i to agent j refers to a sequence of directed edges connecting agents i to j. A directed graph is strongly connected if any agent is reachable from all other agents via a directed path. A family of directed graphs $\{\mathcal{G}_p\}_{p=1}^M$ is said to be jointly strongly connected if their union graph $\mathcal{G}_U = \bigcup_{p=1}^M \mathcal{G}_p$ is strongly connected.

Assumption 1: Each candidate directed topology $\mathcal{G}_p$ is disconnected, whereas the family $\{\mathcal{G}_p\}_{p=1}^M$ is jointly strongly connected.

Under the active topology $\mathcal{G}_{\sigma(t)}$, where $\sigma(t) \in \Omega = \{1,\ldots,M\}$ is the switching signal, the agent dynamics under the standard consensus protocol are given by

$$\dot{x}_i(t) = -\sum_{j=1}^{N} a^{\sigma(t)}_{ij}\big(x_i(t) - x_j(t)\big). \tag{1}$$

By stacking $x(t) = \mathrm{col}\{x_1(t),\ldots,x_N(t)\}$, then (1) can be written in compact form as

$$\dot{x}(t) = -\big(L_{\sigma(t)} \otimes I_d\big)x(t), \tag{2}$$

where I_d is the d-dimensional identity matrix.

The multi-agent system achieves consensus if $\lim_{t\to\infty}\big(x_i(t) - x_j(t)\big) = \mathbf{0}_d$ for all $i,j \in \{1,2,\ldots,N\}$, or equivalently when $x(t)$ converges to the consensus subspace $\mathcal{S}_c = \{\mathbf{1}_N \otimes \eta : \eta \in \mathbb{R}^d\}$.

To describe the disagreement among agents, define the centering matrix $H = I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top$. Let $S \in \mathbb{R}^{(N-1)\times N}$ satisfy $S\mathbf{1}_N = \mathbf{0}_{N-1}$, $SS^\top = I_{N-1}$, $S^\top S = H$. Consequently, the rows of S form an orthonormal basis of the subspace orthogonal to the consensus direction $\mathbf{1}_N$.

Define the reduced-order disagreement vector $\xi(t) = (S \otimes I_d)x(t) \in \mathbb{R}^r$, $r = (N-1)d$. Since $\mathrm{rank}(S) = N-1$ and $S\mathbf{1}_N = \mathbf{0}_{N-1}$, we have $\ker(S) = \mathrm{span}\{\mathbf{1}_N\}$, which implies that $\xi(t) = \mathbf{0}_r$ if and only if there exists some $\eta(t) \in \mathbb{R}^d$ such that $x(t) = \mathbf{1}_N \otimes \eta(t)$. Hence, $\xi(t) \to \mathbf{0}_r$ is equivalent to the original multi-agent system achieving asymptotic consensus.

Define the average state vector $\eta(t) = \frac{1}{N}(\mathbf{1}_N^\top \otimes I_d)x(t) \in \mathbb{R}^d$. Using $I_N = \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top + S^\top S$, the original state can be decomposed as

$$x(t) = \mathbf{1}_N \otimes \eta(t) + (S^\top \otimes I_d)\xi(t). \tag{3}$$

Differentiating $\xi(t)$ and using (2) and (3) gives

$$\begin{aligned}\dot{\xi}(t) &= -(SL_{\sigma(t)} \otimes I_d)x(t)\\ &= -\big(SL_{\sigma(t)}\mathbf{1}_N \otimes I_d\big)\eta(t) - \big(SL_{\sigma(t)}S^\top \otimes I_d\big)\xi(t)\\ &= -\big(SL_{\sigma(t)}S^\top \otimes I_d\big)\xi(t).\end{aligned} \tag{4}$$

Define $\bar{A}_p = -\big(SL_pS^\top \otimes I_d\big)$. Then, the reduced-order disagreement dynamics are represented by the switched linear system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t). \tag{5}$$

The objective of this paper is to design a state-dependent dwell-time switching law for the agents' communication network such that consensus can be achieved even when all candidate communication topologies are disconnected.

III. State-Dependent Dwell-Time Switching Design for the Reduced Linear System

Consider the reduced-order disagreement system (5). Define the disagreement output as $z(t) = C\xi(t) \in \mathbb{R}^q$, where $C \in \mathbb{R}^{q\times r}$, $q \geq r$, is a given full-column-rank matrix. In particular, $C = I_r$ corresponds to $q = r$ and gives direct measurement of the complete disagreement state.

For each mode $i \in \Omega$, choose a symmetric positive-definite matrix $X_i = X_i^\top \in \mathbb{R}^{r\times r} \succ 0$ and define $V_i(\xi) = \xi^\top X_i \xi$. Let $\Pi = [\pi_{ij}] \in \mathcal{M}$ be a Metzler matrix satisfying $\pi_{ij} \geq 0$, for $i \neq j$, and $\sum_{j=1}^{M} \pi_{ij} = 0$. For a prescribed minimum dwell time $T > 0$, define

$$Y_{1,j} = e^{\bar{A}_j^\top T} X_j e^{\bar{A}_j T},\ Y_{2,j} = \int_0^T e^{\bar{A}_j^\top \tau} C^\top C e^{\bar{A}_j \tau}\, d\tau. \quad (6)$$

The finite-dwell-time Lyapunov-Metzler inequalities are given by

$$\bar{A}_i^\top X_i + X_i \bar{A}_i + \sum_{j\neq i} \pi_{ij} \left(Y_{1,j} + Y_{2,j} - X_i\right) + C^\top C \prec 0. \quad (7)$$

For a given active mode $i = \sigma(t_k)$, define $\Omega_i = \Omega \setminus \{i\}$. For every $j \in \Omega_i$, let

$$\begin{aligned} \Delta_{ij}(t) &= \xi^\top(t)\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t), \\ J_j(t) &= \xi^\top(t)\left(Y_{1,j} + Y_{2,j}\right)\xi(t). \end{aligned} \quad (8)$$

The next switching instant is defined as

$$t_{k+1} = \inf\left\{t > t_k + T : \exists j \in \Omega_i,\ \Delta_{ij}(t) < 0\right\}, \quad (9)$$

and the active topology is selected according to

$$\begin{aligned} \sigma(t) &= i, & t \in [t_k, t_{k+1}), \\ \sigma(t_{k+1}) &= \arg\min_{j\in\Omega_i} J_j(t_{k+1}). \end{aligned} \quad (10)$$

By construction, the switching law satisfies $t_{k+1} - t_k \geq T$, and therefore excludes Zeno behavior.

Based on the above switching law, the following result follows directly by applying the state-dependent dwell-time stabilization framework in [19] to the reduced-order disagreement system (5).

Theorem 1: Suppose that there exist matrices $X_i = X_i^\top \succ 0$, $i \in \Omega$, and a Metzler matrix $\Pi \in \mathcal{M}$ satisfying the Lyapunov-Metzler inequalities (7). Then the switching law (9)–(10) renders the disagreement system (5) asymptotically stable. In particular, $\lim_{t\to\infty} \xi(t) = \mathbf{0}_r$. Consequently, the original multi-agent system achieves consensus, that is,

$$\lim_{t\to\infty} \bigl(x_i(t) - x_j(t)\bigr) = \mathbf{0}_d, \qquad \forall i, j \in \{1, \ldots, N\}.$$

The validity of the above theorem follows directly from [19, Theorem 1], and is therefore omitted here.

Remark 1: The inequalities in (7) do not require the individual matrices $\bar{A}_i$ to be Hurwitz; consensus is achieved through state-dependent switching among the candidate topologies.

IV. Existence and Construction of a Metzler Matrix

This section shows that, under Assumption 1, if $Q = C^\top C \succ 0$ and the prescribed minimum dwell time T is sufficiently small, a Metzler matrix can be constructed such that the finite-dwell-time Lyapunov-Metzler inequalities (7) are feasible.

Choose any strictly positive weight vector $\lambda = [\lambda_1, \ldots, \lambda_M]^\top \in \mathbb{R}^M$ with $\lambda_p > 0\ \forall p = 1, \ldots, M$ and $\sum_{p=1}^{M} \lambda_p = 1$. Define $L_\lambda = \sum_{p=1}^{M} \lambda_p L_p$, we obtain

$$\bar{A}_\lambda = \sum_{p=1}^{M} \lambda_p \bar{A}_p = -\bigl(S L_\lambda S^\top \otimes I_d\bigr).$$

Lemma 1: Under Assumption 1, $\bar{A}_\lambda$ is Hurwitz for every strictly positive weight vector λ.

Proof: Since $\lambda_p > 0$ for all p and all adjacency weights are nonnegative, Assumption 1 ensures that the graph associated with L_λ is strongly connected.

Define $\mathcal{T} = \begin{bmatrix}\mathbf{1}_N & S^\top\end{bmatrix}$, $\mathcal{T}^{-1} = \begin{bmatrix}\frac{1}{N}\mathbf{1}_N^\top \\ S\end{bmatrix}$. Using $L_\lambda \mathbf{1}_N = \mathbf{0}_N$, one obtains $\mathcal{T}^{-1} L_\lambda \mathcal{T} = \begin{bmatrix} 0 & \frac{1}{N}\mathbf{1}_N^\top L_\lambda S^\top \\ \mathbf{0}_{N-1} & S L_\lambda S^\top \end{bmatrix}$. Therefore,

$$\operatorname{spec}(L_\lambda) = \operatorname{spec}(\mathcal{T}^{-1} L_\lambda \mathcal{T}) = \{0\} \cup \operatorname{spec}(S L_\lambda S^\top).$$

Since the graph associated with L_λ is strongly connected, the zero eigenvalue of L_λ is simple and all its remaining eigenvalues have strictly positive real parts. Hence, all eigenvalues of $S L_\lambda S^\top$ have strictly positive real parts. Consequently,

$$\bar{A}_\lambda = -\bigl(S L_\lambda S^\top \otimes I_d\bigr)$$

is Hurwitz. ■

For each mode $i \in \Omega$, define $E_i(T) = e^{\bar{A}_i T}$ and $\mathcal{B}_i = \bar{A}_i^\top \oplus \bar{A}_i^\top = \bar{A}_i^\top \otimes I_r + I_r \otimes \bar{A}_i^\top \in \mathbb{R}^{r^2\times r^2}$. Furthermore, let $\mathscr{B} = \operatorname{diag}\{\mathcal{B}_1, \ldots, \mathcal{B}_M\}$, and $\Pi_d = \operatorname{diag}\{\pi_{11}, \ldots, \pi_{MM}\}$. For a matrix tuple $\mathbf{X} = (X_1, \ldots, X_M)$, define the homogeneous linear operator $\mathfrak{F}_T$ componentwise by

$$\mathfrak{F}_T(\mathbf{X})t_i = \bar{A}_i^\top X_i + X_i \bar{A}_i + \pi_{ii} X_i + \sum_{j\neq i} \pi_{ij} E_j^\top(T) X_j E_j(T). \quad (11)$$

Using $\operatorname{vec}(AXB) = (B^\top \otimes A)\operatorname{vec}(X)$ and $E_j^\top(T) \otimes E_j^\top(T) = e^{\bar{A}_j^\top T} \otimes e^{\bar{A}_j^\top T} = e^{\mathcal{B}_j T}$, one obtains

$$\begin{aligned} &\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{X})]_i\right) \\ &= \left[I_r \otimes \bar{A}_i^\top + \bar{A}_i^\top \otimes I_r + \pi_{ii} I_{r^2}\right]\operatorname{vec}(X_i) \\ &\quad + \sum_{j\neq i} \pi_{ij}\left(E_j^\top(T) \otimes E_j^\top(T)\right)\operatorname{vec}(X_j) \\ &= \left(\mathcal{B}_i + \pi_{ii} I_{r^2}\right)\operatorname{vec}(X_i) + \sum_{j\neq i} \pi_{ij} e^{\mathcal{B}_j T}\operatorname{vec}(X_j). \end{aligned} \quad (12)$$

Stacking (12) for $i = 1, \ldots, M$ yields

$$\operatorname{col}\left\{\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{X})]_i\right)\right\}_{i=1}^{M} = \mathcal{J}_T(\Pi)\operatorname{col}\left\{\operatorname{vec}(X_i)\right\}_{i=1}^{M}, \quad (13)$$

where the (i, j)th block of $\mathcal{J}_T$ is

$$[\mathcal{J}_T(\Pi)]_{ij} = \begin{cases} \mathcal{B}_i + \pi_{ii} I_{r^2}, & i = j, \\ \pi_{ij} e^{\mathcal{B}_j T}, & i \neq j. \end{cases}$$

Since $e^{\mathscr{B}T} = \operatorname{diag}\left\{e^{\mathcal{B}_1 T}, \ldots, e^{\mathcal{B}_M T}\right\}$, the above block matrix can be written compactly as

$$\mathcal{J}_T(\Pi) = \mathscr{B} + \Pi_d \otimes I_{r^2} + \left[(\Pi - \Pi_d) \otimes I_{r^2}\right] e^{\mathscr{B}T}. \quad (14)$$

Lemma 2: Given a Metzler matrix $\Pi \in \mathcal{M}$, a prescribed minimum dwell time $T > 0$, and a matrix C satisfying $Q = C^\top C \succ 0$, define $\mathcal{J}_T(\Pi)$ as in (14). If $\mathcal{J}_T(\Pi)$ is Hurwitz, then the Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions

$$X_i = X_i^\top \succ 0, \qquad i = 1, \ldots, M.$$

Proof: Define $W_i(T) = Q + \sum_{j \neq i} \pi_{ij} Y_{2,j}(T)$. Since $Y_{2,j}(T) = \int_0^T e^{\bar{A}_j^\top \tau} Q e^{\bar{A}_j \tau} d\tau \succ 0$, it follows that $W_i(T) \succ 0$. Using $\pi_{ii} = -\sum_{j \neq i} \pi_{ij}$, the Lyapunov-Metzler inequalities can be written as

$$[\mathfrak{F}_T(\mathbf{X})]_i + W_i(T) \prec 0, \qquad i = 1, \ldots, M. \tag{15}$$

Suppose that $\mathcal{J}_T(\Pi)$ is Hurwitz. Consider the auxiliary matrix-valued linear system

$$\dot{\mathbf{Z}}(s) = \mathfrak{F}_T\big(\mathbf{Z}(s)\big), \mathbf{Z}(s) = \big(Z_1(s), \ldots, Z_M(s)\big) \in (\mathbb{S}^r)^M. \tag{16}$$

Since $\mathfrak{F}_T$ maps symmetric matrix tuples into symmetric matrix tuples, $(\mathbb{S}^r)^M$ is invariant under (16).

Define the stacked vector $\boldsymbol{\zeta}(s) = \operatorname{col}\{\operatorname{vec}\big(Z_i(s)\big)\}_{i=1}^M \in \mathbb{R}^{Mr^2}$. By (13), system (16) is equivalent to

$$\dot{\boldsymbol{\zeta}}(s) = \mathcal{J}_T(\Pi)\boldsymbol{\zeta}(s).$$

Hence, $\boldsymbol{\zeta}(s) = e^{\mathcal{J}_T(\Pi)s}\boldsymbol{\zeta}(0)$. Since $\mathcal{J}_T(\Pi)$ is Hurwitz, this finite-dimensional linear system is exponentially stable. Therefore, there exist constants $c > 0$ and $\gamma > 0$ such that

$$\|\boldsymbol{\zeta}(s)\|_2 \leq ce^{-\gamma s}\|\boldsymbol{\zeta}(0)\|_2, \qquad s \geq 0. \tag{17}$$

In particular, $\boldsymbol{\zeta}(s) \to \mathbf{0}_{Mr^2}$ as $s \to \infty$, and therefore $Z_i(s) \to \mathbf{0}_{r \times r}$ for all $i = 1, \ldots, M$.

Define $\widetilde{A}_i = \bar{A}_i + \frac{\pi_{ii}}{2} I_r$. Then the ith component of (16) is

$$\dot{Z}_i = \widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i + \sum_{j \neq i} \pi_{ij} E_j^\top(T) Z_j E_j(T). \tag{18}$$

We now verify the tangent-cone condition for $(\mathbb{S}_+^r)^M$. Let $\mathbf{Z} = (Z_1, \ldots, Z_M) \in (\mathbb{S}_+^r)^M$ and take any $v \in \ker(Z_i) \subset \mathbb{R}^r$. Since $Z_i v = \mathbf{0}_r$,

$$v^\top \dot{Z}_i v = \sum_{j \neq i} \pi_{ij} \big(E_j(T)v\big)^\top Z_j \big(E_j(T)v\big) \geq 0, \tag{19}$$

where the inequality follows from $\pi_{ij} \geq 0$ and $Z_j \succeq 0$.

According to [20], the tangent cone of $\mathbb{S}_+^r$ at Z_i is given by

$$T_{\mathbb{S}_+^r}(Z_i) = \left\{H \in \mathbb{S}^r : v^\top H v \geq 0, \ \forall v \in \ker(Z_i)\right\}.$$

Hence, by (19) and the tangent-cone rule for Cartesian products,

$$\mathfrak{F}_T(\mathbf{Z}) \in T_{(\mathbb{S}_+^r)^M}(\mathbf{Z}), \qquad \forall \mathbf{Z} \in (\mathbb{S}_+^r)^M.$$

Since $(\mathbb{S}_+^r)^M$ is a closed convex cone, the Nagumo invariance theorem [21] implies

$$\mathbf{Z}(0) \in (\mathbb{S}_+^r)^M \implies \mathbf{Z}(s) \in (\mathbb{S}_+^r)^M, \qquad s \geq 0.$$

Choose an arbitrary scalar $\rho > 1$ and define $R_i = \rho W_i(T) \succ 0$. Let $\mathbf{R} = (R_1, \ldots, R_M)$ and let $\mathbf{Z}(s; \mathbf{R}) = \big(Z_1(s; \mathbf{R}), \ldots, Z_M(s; \mathbf{R})\big)$ denote the solution of (16) with $\mathbf{Z}(0; \mathbf{R}) = \mathbf{R}$. Define $\mathbf{r} = \operatorname{col}\{\operatorname{vec}(R_i)\}_{i=1}^M \in \mathbb{R}^{Mr^2}$. The associated stacked solution satisfies $\operatorname{col}\{\operatorname{vec}\big(Z_i(s; \mathbf{R})\big)\}_{i=1}^M = e^{\mathcal{J}_T(\Pi)s}\mathbf{r}$. Therefore, by (17),

$$\left\|\operatorname{col}\{\operatorname{vec}\big(Z_i(s; \mathbf{R})\big)\}_{i=1}^M\right\|_2 \leq ce^{-\gamma s}\|\mathbf{r}\|_2.$$

Hence, every component $Z_i(s; \mathbf{R})$ is integrable on $[0, \infty)$.

For each $i \in \Omega$, define

$$X_i = \int_0^\infty Z_i(s; \mathbf{R})\, ds, \tag{20}$$

and let $\mathbf{X} = (X_1, \ldots, X_M)$. Since $\mathbf{R} \in (\mathbb{S}_+^r)^M$ and the cone $(\mathbb{S}_+^r)^M$ is forward invariant, $Z_i(s; \mathbf{R}) \succeq 0, s \geq 0$. Therefore,

$$X_i = \int_0^\infty Z_i(s; \mathbf{R})\, ds \succeq 0, \qquad i = 1, \ldots, M.$$

In particular, $X_i = X_i^\top$.

To establish strict positive definiteness, observe that $Z_i(0; \mathbf{R}) = R_i = \rho W_i(T) \succ 0$. Since $Z_i(s; \mathbf{R})$ is continuous with respect to s, there exists $\delta_i > 0$ such that $Z_i(s; \mathbf{R}) \succ 0, 0 \leq s \leq \delta_i$. Consequently,

$$X_i = \int_0^\infty Z_i(s; \mathbf{R})\, ds \ \succeq \int_0^{\delta_i} Z_i(s; \mathbf{R})\, ds \ \succ 0.$$

Thus,

$$X_i = X_i^\top \succ 0.$$

Finally, since $\mathfrak{F}_T$ is a finite-dimensional linear operator and the above integral is convergent, $\mathfrak{F}_T$ can be moved inside the integral. Using $\dot{\mathbf{Z}}(s; \mathbf{R}) = \mathfrak{F}_T\big(\mathbf{Z}(s; \mathbf{R})\big)$, one obtains

$$\begin{aligned}
\mathfrak{F}_T(\mathbf{X}) &= \mathfrak{F}_T\left(\int_0^\infty \mathbf{Z}(s; \mathbf{R})\, ds\right) \\
&= \int_0^\infty \mathfrak{F}_T\big(\mathbf{Z}(s; \mathbf{R})\big) ds \\
&= \int_0^\infty \dot{\mathbf{Z}}(s; \mathbf{R}) ds \\
&= \lim_{s \to \infty} \mathbf{Z}(s; \mathbf{R}) - \mathbf{Z}(0; \mathbf{R}) \\
&= -\mathbf{R}.
\end{aligned}$$

Therefore, for every $i = 1, \ldots, M$,

$$\begin{aligned}
\mathfrak{F}_T(\mathbf{X})t_i + W_i(T) &= -R_i + W_i(T) \\
&= -\rho W_i(T) + W_i(T) \\
&\prec 0.
\end{aligned}$$

Hence, the Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions. ■

Define $\Pi_0 = \mathbf{1}_M \lambda^\top - I_M$, $\Pi_\kappa = \kappa \Pi_0$, $\kappa > 0$. Since $\lambda > 0$ entrywise, Π_κ is Metzler. Using $\lambda^\top \mathbf{1}_M = 1$, it follows that

$$\begin{aligned}
\Pi_\kappa \mathbf{1}_M &= \kappa\big(\mathbf{1}_M \lambda^\top \mathbf{1}_M - \mathbf{1}_M\big) = \mathbf{0}_M, \\
\lambda^\top \Pi_\kappa &= \kappa\big(\lambda^\top \mathbf{1}_M \lambda^\top - \lambda^\top\big) = \mathbf{0}_M^\top.
\end{aligned}$$

For $T = 0$, $e^{\mathscr{B}T} = I_{Mr^2}$, and hence (14) simplifies to

$$\mathcal{J}_0(\Pi_\kappa) = \mathscr{B} + \Pi_\kappa \otimes I_{r^2}. \tag{21}$$

Lemma 3: Under Assumption 1, there exists $\kappa^\star > 0$ such that $\mathcal{J}_0(\Pi_\kappa)$ is Hurwitz for every $\kappa > \kappa^\star$.

Proof: Π_0 has a simple zero eigenvalue with right eigenvector $\mathbf{1}_M$ and left eigenvector λ, satisfying $\Pi_0 \mathbf{1}_M = \mathbf{0}_M$ and $\lambda^\top \Pi_0 = \mathbf{0}_M^\top$. Furthermore, for every $y \in \ker(\lambda^\top) \subset \mathbb{R}^M$, $\Pi_0 y = \mathbf{1}_M(\lambda^\top y) - y = -y$. Since $\dim \ker(\lambda^\top) = M - 1$, one obtains $\sigma(\Pi_0) = \left\{0, \underbrace{-1, \ldots, -1}_{M-1 \text{ times}}\right\}$.

Choose $\Phi \in \mathbb{R}^{M \times (M-1)}$ whose columns form a basis of $\ker(\lambda^\top)$, and define $U_M = \begin{bmatrix}\mathbf{1}_M & \Phi\end{bmatrix}$. Since $\lambda^\top \mathbf{1}_M = 1$ and

$\lambda^\top \Phi = \mathbf{0}_{M-1}^\top$, the matrix U_M is nonsingular and the first row of U_M^{-1} is $\lambda^\top$. It follows that

$$\Pi_0 U_M = \begin{bmatrix} \Pi_0 \mathbf{1}_M & \Pi_0 \Phi \end{bmatrix} = \begin{bmatrix} \mathbf{0}_M & -\Phi \end{bmatrix} = U_M \begin{bmatrix} 0 & \mathbf{0}_{M-1}^\top \\ \mathbf{0}_{(M-1)} & -I_{M-1} \end{bmatrix}.$$

which yields $U_M^{-1} \Pi_0 U_M = \mathrm{diag}\{0, -I_{M-1}\}$.

Define $U = U_M \otimes I_{r^2}$. Partition $U^{-1} \mathscr{B} U = \begin{bmatrix} \bar{\mathcal{B}} & \mathcal{B}_{12} \\ \mathcal{B}_{21} & \mathcal{B}_{22} \end{bmatrix}$, where $\bar{\mathcal{B}} = (\lambda^\top \otimes I_{r^2}) \mathscr{B} (\mathbf{1}_M \otimes I_{r^2}) = \sum_{i=1}^{M} \lambda_i \mathcal{B}_i$. Moreover, $U^{-1} (\Pi_\kappa \otimes I_{r^2}) U = \mathrm{diag}\left\{0, -\kappa I_{(M-1)r^2}\right\}$. Therefore,

$$U^{-1} \mathcal{J}_0(\Pi_\kappa) U = \begin{bmatrix} \bar{\mathcal{B}} & \mathcal{B}_{12} \\ \mathcal{B}_{21} & \mathcal{B}_{22} - \kappa I_{(M-1)r^2} \end{bmatrix}. \quad (22)$$

Using $\mathcal{B}_i = \bar{A}_i^\top \oplus \bar{A}_i^\top$, one obtains

$$\bar{\mathcal{B}} = \sum_{i=1}^{M} \lambda_i \left(\bar{A}_i^\top \oplus \bar{A}_i^\top \right) = \bar{A}_\lambda^\top \oplus \bar{A}_\lambda^\top.$$

By Lemma 1, $\bar{A}_\lambda$ is Hurwitz. Therefore, $\bar{\mathcal{B}}$ is Hurwitz.

Let $P_{\mathcal{B}} = P_{\mathcal{B}}^\top \succ 0$ be the unique solution of $\bar{\mathcal{B}}^\top P_{\mathcal{B}} + P_{\mathcal{B}} \bar{\mathcal{B}} = -I_{r^2}$. Define $\mathcal{P} = \mathrm{diag}\left\{P_{\mathcal{B}}, I_{(M-1)r^2}\right\} \succ 0$ and $\mathcal{H} = P_{\mathcal{B}} \mathcal{B}_{12} + \mathcal{B}_{21}^\top$. Denote the transformed matrix in (22) by $\widehat{\mathcal{J}}_0(\Pi_\kappa)$. Then

$$\widehat{\mathcal{J}}_0^\top(\Pi_\kappa) \mathcal{P} + \mathcal{P} \widehat{\mathcal{J}}_0(\Pi_\kappa) = \begin{bmatrix} -I_{r^2} & \mathcal{H} \\ \mathcal{H}^\top & \mathcal{B}_{22}^\top + \mathcal{B}_{22} - 2\kappa I_{(M-1)r^2} \end{bmatrix}.$$

Choose $\kappa^\star > 0$ sufficiently large such that

$$2\kappa^\star > \lambda_{\max} \left(\mathcal{B}_{22}^\top + \mathcal{B}_{22} + \mathcal{H}^\top \mathcal{H} \right).$$

Then, for every $\kappa > \kappa^\star$,

$$\mathcal{B}_{22}^\top + \mathcal{B}_{22} - 2\kappa I_{(M-1)r^2} + \mathcal{H}^\top \mathcal{H} \prec 0.$$

Since the upper-left block is $-I_{r^2} \prec 0$, the Schur-complement condition gives

$$\widehat{\mathcal{J}}_0^\top(\Pi_\kappa) \mathcal{P} + \mathcal{P} \widehat{\mathcal{J}}_0(\Pi_\kappa) \prec 0.$$

Hence, $\widehat{\mathcal{J}}_0(\Pi_\kappa)$ is Hurwitz. Because $\widehat{\mathcal{J}}_0(\Pi_\kappa)$ is similar to $\mathcal{J}_0(\Pi_\kappa)$, it follows that $\mathcal{J}_0(\Pi_\kappa)$ is Hurwitz for every $\kappa > \kappa^\star$. ∎

Theorem 2: Suppose that Assumption 1 holds and $Q = C^\top C \succ 0$. Choose a strictly positive convex vector $\lambda = [\lambda_1, \ldots, \lambda_M]^\top$, $\sum_{i=1}^M \lambda_i = 1$, $\lambda_i > 0$. Let $\kappa > \kappa^\star$, where $\kappa^\star$ is given by Lemma 3, and set $\Pi = \kappa \left(\mathbf{1}_M \lambda^\top - I_M \right)$. Let $\mathcal{J}_0(\Pi) = \mathscr{B} + \Pi \otimes I_{r^2}$, and let $P = P^\top \succ 0$ be the unique solution of $\mathcal{J}_0(\Pi)^\top P + P \mathcal{J}_0(\Pi) = -I_{Mr^2}$. Define

$$T_{\mathrm{ub}} = \frac{1}{\|\mathscr{B}\|_2} \ln \left(1 + \frac{1}{2\|P\|_2 \|\Pi - \Pi_d\|_2} \right). \quad (23)$$

Then, for every prescribed dwell time satisfying $0 < T < T_{\mathrm{ub}}$, the finite-dwell-time Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions

$$X_i = X_i^\top \succ 0, \qquad i = 1, \ldots, M.$$

Proof: Let $\Delta_T = \mathcal{J}_T(\Pi) - \mathcal{J}_0(\Pi)$. By (14) and the definition of $\mathcal{J}_0(\Pi)$, $\Delta_T = [(\Pi - \Pi_d) \otimes I_{r^2}] \left(e^{\mathscr{B}T} - I_{Mr^2} \right)$.

By the submultiplicativity of the spectral norm and the identity $\|A \otimes B\|_2 = \|A\|_2 \|B\|_2$,

$$\begin{aligned} \|\Delta_T\|_2 &\le \|(\Pi - \Pi_d) \otimes I_{r^2}\|_2 \|e^{\mathscr{B}T} - I_{Mr^2}\|_2 \\ &= \|\Pi - \Pi_d\|_2 \|e^{\mathscr{B}T} - I_{Mr^2}\|_2. \end{aligned} \quad (24)$$

Using $e^{\mathscr{B}T} - I_{Mr^2} = \int_0^T \mathscr{B} e^{\mathscr{B}s} ds$, we derive

$$\|e^{\mathscr{B}T} - I_{Mr^2}\|_2 \le \int_0^T \|\mathscr{B}\|_2 e^{\|\mathscr{B}\|_2 s} ds = e^{\|\mathscr{B}\|_2 T} - 1. \quad (25)$$

Combining (24) and (25) yields

$$\|\Delta_T\|_2 \le \|\Pi - \Pi_d\|_2 \left(e^{\|\mathscr{B}\|_2 T} - 1 \right). \quad (26)$$

Since $\mathcal{J}_T(\Pi) = \mathcal{J}_0(\Pi) + \Delta_T$,

$$\begin{aligned} \mathcal{J}_T(\Pi)^\top P + P \mathcal{J}_T(\Pi) &= \mathcal{J}_0(\Pi)^\top P + P \mathcal{J}_0(\Pi) + \Delta_T^\top P + P \Delta_T \\ &= -I_{Mr^2} + \Delta_T^\top P + P \Delta_T. \end{aligned} \quad (27)$$

For any $y \in \mathbb{R}^{Mr^2}$, $y^\top (\Delta_T^\top P + P \Delta_T) y = 2 y^\top P \Delta_T y \le 2\|P\|_2 \|\Delta_T\|_2 \|y\|_2^2$, thus

$$\Delta_T^\top P + P \Delta_T \preceq 2\|P\|_2 \|\Delta_T\|_2 I_{Mr^2}.$$

Substituting into (27) yields

$$\mathcal{J}_T(\Pi)^\top P + P \mathcal{J}_T(\Pi) \preceq -\left[1 - 2\|P\|_2 \|\Delta_T\|_2\right] I_{Mr^2}. \quad (28)$$

A sufficient condition for $\mathcal{J}_T(\Pi)$ to be Hurwitz is $2\|P\|_2 \|\Delta_T\|_2 < 1$. By (26), this is guaranteed by

$$2\|P\|_2 \|\Pi - \Pi_d\|_2 \left(e^{\|\mathscr{B}\|_2 T} - 1 \right) < 1,$$

which rearranges to

$$T < \frac{1}{\|\mathscr{B}\|_2} \ln \left(1 + \frac{1}{2\|P\|_2 \|\Pi - \Pi_d\|_2} \right) \triangleq T_{\mathrm{ub}}.$$

Thus, for all $0 < T < T_{\mathrm{ub}}$, $\mathcal{J}_T(\Pi)^\top P + P \mathcal{J}_T(\Pi) \prec 0$. By the Lyapunov stability theorem, $\mathcal{J}_T(\Pi)$ is Hurwitz. By Lemma 2, the finite-dwell-time Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions. ∎

V. Numerical Simulations

Consider a multi-agent system with $N = 4$ agents, $d = 1$, and two candidate topologies shown in Fig. 1. Each topology is disconnected, whereas the two candidate topologies are jointly strongly connected.

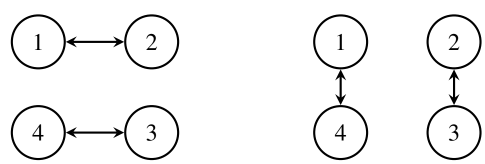

Fig. 1. Candidate communication topologies.

The corresponding reduced matrices satisfy $\sigma(\bar{A}_1) = \sigma(\bar{A}_2) = \{-2, -2, 0\}$, and hence neither subsystem is Hurwitz. For $\lambda = [0.5, 0.5]^\top$, the convex combination satisfies $\sigma(\bar{A}_\lambda) = \{-2, -1, -1\}$, which verifies Lemma 1.

Choose $\Pi = \begin{bmatrix} -0.27 & 0.27 \\ 0.27 & -0.27 \end{bmatrix}$, $C = I_3$. Theorem 2 gives $T_{\mathrm{ub}} = 0.1647$ s. Accordingly, the prescribed dwell time is selected as $T = 0.1$ s $< T_{\mathrm{ub}}$, for which the Lyapunov-Metzler inequalities are feasible.

For $x(0) = [5, -3, 4, 2]^\top$, $\sigma(0) = 1$, the simulation results are shown in Figs. 2 and 3.

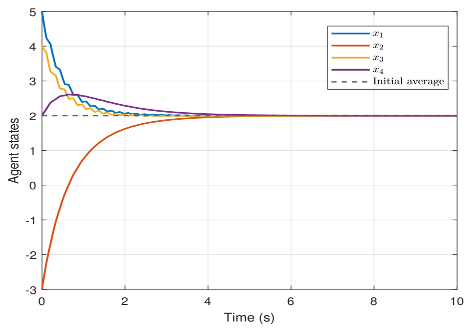

Fig. 2. Agent state trajectories under the proposed switching law.

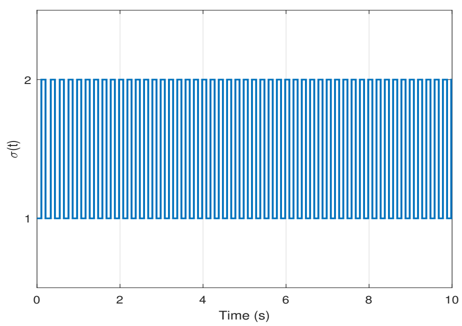

Fig. 3. State-dependent switching signal.

The agent states converge to the common value 2, while every switching interval is no less than the prescribed dwell time $T = 0.1$ s. Therefore, consensus is achieved although neither candidate topology can independently guarantee global consensus.

VI. Conclusion

This paper studied the consensus problem for multi-agent systems under switching directed topologies. A consensus-orthogonal decomposition was used to transform the original problem into stabilization of a reduced-order switched disagreement system. A state-dependent switching law with a prescribed minimum dwell time was developed based on Lyapunov-Metzler inequalities. Under the joint strong connectivity of the candidate topology family, a Hurwitz convex combination of the reduced subsystem matrices was obtained and used to construct a Metzler matrix. The feasibility of the Lyapunov-Metzler inequalities were established and an explicit sufficient upper bound on the dwell time was derived. Numerical results demonstrated that consensus can be achieved even when none of the individual subsystem matrices is Hurwitz.

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